

Vehicle Cybersecurity & Hardware Security Module Test Report

ISO 21434, Secure Boot, SecOC, JTAG Inspection, and TRNG Entropy

1. Hardware Security Module Secure Boot Benchmarks

- Test Case TC-SEC-081: Secure boot benchmark: RSA-3072 signature verification completed in 112.4 ms with valid public key. Verdict

2. AUTOSAR SecOC Hardware CMAC Acceleration

- Test Case TC-SEC-082: SecOC hardware acceleration benchmark: measured AES-128 CMAC generation time was 28.5 μs per mess

3. Keystore Hardware Isolation & IDS Rate Limiter

- Test Case TC-SEC-083: AES symmetric root keys verified inside HSM secure storage. Asymmetric ECC private key migration to secur
- Test Case TC-SEC-084: IDS rate limiter evaluated on CAN 1 (Powertrain) and CAN 2 (Body); blocked injection in 32.0 ms. CAN 3 (Ch

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4. Hardware Vulnerability Inspection & TRNG Entropy Benchmarks

- Inspection Finding SEC-085: Hardware inspection: JTAG port pins remain accessible with debugging enabled on production sample lo
- Test Case TC-SEC-086: NIST SP 800-22 random number test suite: measured TRNG output rate was 140.0 kbps, failing the 500.0 kb